

# **ADVANCED AUTONOMOUS SYSTEMS DESIGN AND OPERATION COMPETITION DETAILED EVALUATION REPORT PREPARATION INSTRUCTION**

**TEAM NAME:**

**TEAM ID:**

**APPLICATION ID:**

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## VERSIONS

*Table. Versions*

| DATE | DATE       | DESCRIPTION                  |
|------|------------|------------------------------|
| V 1  | 14.05.2026 | TEKNOFEST 2026 First Version |

## DEFINITIONS and ABBREVIATIONS

The abbreviations used within the scope of this specification and their corresponding definitions are provided in this section.

ARB: Advisory and Review Board

UAV: Unmanned Aerial Vehicle

UGV: Unmanned Ground Vehicle

TDZ: Target Detection Zone

UAV-LZ: UAV Launch Zone

UAV-OZ: UAV Operation Zone

UGV-PZ: UGV Positioning Zone

TZ: Target Zone

UAV-LZ: UAV Landing Zone

PRR: Preliminary Review Report

DRR: Detailed Review Report

FRR: Final Review Report

FRS: Final Review Stage

FRP: Final Review Presentation

SRP: System Review Presentation

FRRR: Flight Readiness Review Report

Cdifficulty: Difficulty Coefficient

wUAV: UAV Weight Score

dUAV: UAV Mission Duration Score

dpreparation: Preparation Duration Score

wUGV: UGV Weight Score

wpayload: UGV Payload Score

dUGV: UGV Mission Duration Score

Smission: Mission Success

S: Competitor Performance Score

STI: Technical Inspection Performance Score

Sflight: Flight Performance Score

Ssportmanship: Sportsmanship Score

Ddeduction: Deduction Score

## INTERNATIONAL ADVANCED AUTONOMOUS SYSTEMS DESIGN AND OPERATION COMPETITION DETAILED EVALUATION REPORT PREPARATION GUIDELINE

This guide serves as a guide for the Detailed Evaluation Report (DDR) that teams applying to the Advanced Autonomous Systems Design and Operation Competition are obliged to prepare. It is expected that this report, which is the second milestone of the competition schedule, will be prepared with utmost diligence based on the guidelines and technical criteria presented under the relevant topic headings. The Detailed Evaluation Report (DDR) will be evaluated according to the table given below in line with the criteria determined by a board possessing technical competence. Successfully completing this stage constitutes a prerequisite for proceeding to the next technical phase.

Table 1. Detailed Evaluation Report Evaluation Scores

| Section No       | Section Name                                  | Points |
|------------------|-----------------------------------------------|--------|
| <b>Section 1</b> | Summary                                       | 10     |
| <b>Section 2</b> | Organizational Structure and Project Schedule | 10     |
| <b>Section 3</b> | General System Specifications                 | 60     |
| <b>Section 4</b> | Test Activities                               | 10     |
| <b>Section 5</b> | Requirements Traceability and Evaluation      | 10     |

### Section 1. Summary (10 POINTS)

Within the scope of this report, a systematic and holistic summary defining the technical outlines and operational capability of the project is required to be provided. The preliminary design activities and original system architectures carried out by taking into consideration the project requirements of the UAV, UGV, and Catapult systems should be defined. By evaluating the configurations belonging to the systems with appropriate evaluation metrics, their technical compatibility with mission scenarios, their endurance under boundary conditions, and the mission flexibility they provide against variable operational conditions should be analyzed, and accordingly, which of the configurations stands out should be defined together with its justifications. Finally, how each of the systems is integrated into the mission definitions within the scope of the competition and the contribution of these systems to operational success should be summarized.

## **Section 2. Organizational Structure and Project Schedule (10 POINTS)**

In order to clarify the institutional functioning and implementation phases of the program, this section examines the organizational hierarchy in which roles are defined and the project schedule covering the stages of the design cycle. These two fundamental elements, which ensure that the engineering cycle progresses in a systematic order, are presented in detail under two separate headings below.

**Section 2.1. Organizational Structure;** The current organizational structure of the team and the execution authorities of the members/units should be schematized in this section. In accordance with the objective evaluation criteria of the report, a role-based and professional description that is completely free of personal data of the members (name, image, etc.) should be preferred.

**Section 2.2. Project Schedule;** Within the scope of this section, compliance with the schedule plan included in the project schedule previously prepared in the Preliminary Evaluation Report should be confirmed, and if there are new workflows added to the work schedule, the planning should be re-evaluated and updated with an analytical approach.

## **Section 3. General System Specifications (60 POINTS)**

In this section, the system specifications defined for the main system and subsystems to be developed within the scope of the competition must be presented in a comprehensive manner. The aforementioned specifications will be examined within the axes of Flight Sciences, Structural Design and Analysis, and Avionics-Software disciplines, and the system's holistic architecture will be addressed under three main headings.

**Section 3.1. Flight Sciences - UAV Conceptual Design Principles;** Under this subsection, the technical specifications of the unmanned aerial vehicle to be designed, including weight-balance data, operational mission profile, aerodynamic structural design processes, stability analyses, and flight performance parameters, shall be addressed in a comprehensive manner.

**Section 3.1.1 Weight & Balance;** In this section, the weight and balance calculations of the unmanned aerial vehicle designed must be explained in all details under the following three headings at the required level of detail.



Weight Estimations: The methods followed in estimating the aircraft weight, the statistical data or empirical formulas used, and the reference points must be explained together with their bases. Any assumptions made and safety margins, if any, must be presented with justification.

Weight and Balance Tables: A detailed component-based mass breakdown table must be created. The location of the center of gravity (CG) on the aircraft must be visualized. Any assumptions or estimations made, if any, must be explained together with their bases.

Final Moment of Inertia Values: The moment of inertia values that will serve as input to dynamic stability analyses must be presented together with the calculation methods. Any assumptions or estimations made, if any, must be explained together with their bases.

**Section 3.1.2 Mission Profile:** The execution of the task defined within the framework of the competition rules shall be divided into flight phases, and the mission profile of the aircraft shall be formed and explained in detail.

**Section 3.1.3 Constraint Diagram:** The constraint diagram of the designed aircraft must be explained and presented in full detail. The methodology followed while creating the constraint diagram must be explained comprehensively. Finally, the selected design point from the constraint diagram must be determined, and the selected values of wing loading (W/S), power loading (W/P), and power-to-weight ratio (P/W) must be presented together with their justification.

**Section 3.1.4 Wing Design:** Under this heading, the methodology followed in wing design must be explained in full detail. All equations used or derived in the application of the methodology must be provided. Input parameters used in the design must be presented together with their estimated bases. The selected airfoil, wing placement, chord lengths, wingspan, and all other parameters of the wing must be specified.

**Section 3.1.5 Fuselage Design:** The methodology followed in the fuselage design of the aircraft must be explained in full detail. All equations used or derived in the application of the methodology must be provided. Input parameters used in the design must be presented together

with their estimated bases. All parameters such as fuselage diameter, length, etc., must be specified.

**Section 3.1.6 Tail Design:** The methodology followed in the tail design of the aircraft must be explained in full detail. All equations used or derived in the application of the methodology must be provided. Input parameters used in the design must be presented together with their estimated bases. All parameters such as the airfoil profiles selected for the tail unit, tail configuration, chord lengths, wingspans, etc., must be specified.

**Section 3.1.7 Control Surface Design:** The methodology followed in the control surface design of the aircraft must be explained in full detail. All equations used or derived in the application of the methodology must be provided. Input parameters used in the design must be presented together with their estimated bases. The maximum deflection angles of the obtained control surfaces and all dimensional parameters must be specified.

**Section 3.1.8 Aircraft Aerodynamic Characteristics:** The aerodynamic characteristics of the relevant aircraft must be explained in full detail. The analysis tools used (CFD, semi-empirical formulas, panel methods, etc.) and the reliability of these tools (mesh independence, validation studies, etc.) must be expressed in a technical manner. The obtained aerodynamic parameters of the aircraft must be presented through tables and graphs.

**Section 3.1.9 Aircraft Stability Characteristics:** The static and dynamic stability characteristics of the designed aircraft must be explained in full detail. The methods and tools used in the stability analyses of the aircraft (if any) must be presented comprehensively.

**Section 3.1.10 Flight Envelope:** The flight envelope of the aircraft must be drawn in accordance with the constraints specified in the requirements, including gust lines. The gust equations from CS-VLA must be used.

**Section 3.1.11 Performance Calculations:** The performance calculations for the aircraft must be explained in full detail. The methods (analytical or numerical) and tools (if any) used in the performance calculations must be presented comprehensively.

**Section 3.1.12 Load Calculations:** The load calculations to be performed for the designed aircraft must be explained in full detail. The CS-VLA document must be used in determining

the load cases/conditions of the aircraft. The methods and tools used in the aircraft load calculations (if any) must be presented in a comprehensive manner.

### **Section 3.2 Structural Design and Structural Analysis**

In this section, the structural design parameters of the Unmanned Aerial Vehicle (UAV), Unmanned Ground Vehicle (UGV), and catapult systems to be developed within the scope of the competition, as well as the structural analysis studies conducted to verify these designs, must be explained in a manner that satisfies the defined technical requirements.

**Section 3.2.1 Unmanned Aerial Vehicle (UAV):** Under this heading, the technical specifications of the platform must be addressed in terms of structural design and structural analysis in a complementary manner. Structural requirements from the provided requirements list should be used as a basis, and if necessary, new requirements may be derived; all requirements defined for this stage must be satisfied. Requirement traceability must be considered, and appropriate verification methods (analysis, test, inspection, similarity, etc.) must be defined.

Within the scope of Structural Design processes:

- The master geometry must be fully designed,
- Preliminary design of all components must be completed,
- Control surfaces must be added,
- Necessary interfaces for motor integration must be defined,
- Placement of all systems within the aircraft must be completed,
- A detailed mass distribution table for the current stage of the aircraft must be prepared,
- The center of gravity location (x, y, z coordinates) must be shown using three-dimensional data,
- All fasteners planned to be used in the aircraft must be defined,
- The assembly sequence of the aircraft must be explained within the concept of modularity,
- A part numbering and revision system must be established,
- A Bill of Materials (BOM) must be prepared.

Within the scope of structural analysis processes:

- All calculations to be performed within structural analysis must be demonstrated either through basic (hand calculations) or finite element method approaches,

- All analyses must be conducted under maximum load conditions,
- A table including the strength-to-weight ratio for different materials that can be used in aircraft design must be prepared,
- Static analyses of the fuselage, wing, and tail components of the aircraft must be performed; all avionics equipment within these components must be defined as point loads,
- A comparative modal mass participation table must be presented for all aircraft components,
- Buckling analysis results must be presented separately as eigenvalue tables, including both positive and negative results,
- Buckling eigenvalues must be calculated under the effect of gravitational acceleration,
- Analysis of hinge connections for control surfaces on the aircraft must be performed,
- Calculations for the required torque to be provided by servo motors, if any, on the aircraft must be performed,
- All materials and material models used in finite element models must be presented in detail,
- All constraints and properties of the mesh structure created in finite element models must be presented in tabular form.

**Section 3.2.2 Unmanned Ground Vehicle (UGV):** Under this heading, the technical specifications of the UGV must be addressed in terms of structural design and structural analysis in a complementary manner. Structural requirements from the provided requirements list should be used as a basis, and if necessary, new requirements may be derived; all requirements defined for this stage must be satisfied. Requirement traceability must be considered, and appropriate verification methods (analysis, test, inspection, similarity, etc.) must be defined.

- If the UGV is to be designed by the participants:
  - o The master geometry must be fully designed,
  - o The preliminary design of all parts of the UGV must be completed, necessary connections must be defined, and the placement of all systems must be carried out,
  - o A detailed mass distribution table for the current stage of the UGV must be prepared,
  - o The center of gravity location (x, y, z coordinates) must be shown using three-dimensional data,
  - o All fasteners on the UGV must be identified; fastener types, materials, and locking methods must be defined,

- o A part numbering and revision system must be established and a Bill of Materials (BOM) must be prepared.
- If the UGV is to be procured as a ready-made platform:
  - o The BOM list of the subsystem must be provided,
  - o All modifications to be performed on the vehicle must be described in detail,
  - o Every part to be replaced must be specified,
  - o If any additional designed components exist, their preliminary design must be completed,
  - o Necessary connections must be defined and the placement of all systems must be carried out.

**Section 3.2.3 Catapult:** Under this heading, the technical specifications of the catapult subsystem must be addressed in terms of structural design and structural analysis in a complementary manner. Structural requirements from the provided requirements list should be used as a basis, and if necessary, new requirements may be derived; all requirements defined for this stage must be satisfied. Requirement traceability must be considered, and appropriate verification methods (analysis, test, inspection, similarity, etc.) must be defined.

- If the catapult is to be designed by the participants:
  - o The preliminary design of all components of the subsystem must be completed, necessary connections must be defined, and the placement of all systems must be carried out,
  - o A detailed mass distribution table must be prepared,
  - o The center of gravity location (x, y, z coordinates) must be shown using three-dimensional data,
  - o All fasteners on the catapult must be identified; fastener types, materials, and locking methods must be defined,
  - o A part numbering and revision system must be established and a Bill of Materials (BOM) must be prepared.
- If the catapult is to be procured as a ready-made system:
  - o The BOM list of the subsystem must be provided,
  - o All modifications to be performed on the vehicle must be described in detail,
  - o Every part to be replaced must be specified,
  - o If any additional designed components exist, their preliminary design must be



completed,

o Necessary connections must be defined and the placement of all systems must be carried out.

- For both cases, the catapult used for launch is expected to maintain its structural integrity after firing.
- For both cases, if a spring system is used in the launch mechanism, the spring constant used within the system must be provided.

Structural analysis processes must be considered for the production of the catapult subsystem by the participants within the scope of competition requirements. In this context:

- The range at which the catapult mechanism can launch the UAV must be explained using energy equations and illustrated with a schematic diagram.

### **Section 3.3 Avionics & Software**

In this section, the system architecture, mission flowcharts, hardware selection, wiring activities, EMI/EMC measures, power systems, and battery calculations of the Unmanned Aerial Vehicle (UAV) and Unmanned Ground Vehicle (UGV) systems to be developed within the scope of the competition, as well as the details of the planned software activities, must be included.

**Section 3.3.1 System Architecture and Mission Flowcharts:** For both the UAV and UGV separately, detailed and readable connection diagrams must be provided, showing how all electronic components are interconnected and distinguishing signal lines (PWM, UART, I2C, etc.) and power lines using color coding. In addition, detailed mission flowcharts for both the UAV and UGV must be provided. Finally, logic diagrams of fail-safe scenarios created in accordance with the requirements must be presented within this section.

**Section 3.3.2 Hardware Selection and Comparative Analysis:** The selection process of all avionics components used in the UAV and UGV platforms must be presented in detail, based on comparative analyses with market alternatives, along with technical justifications and performance advantages of the selected products. In addition, comprehensive Bills of Materials (BOM) for UAV and UGV platforms must be presented separately in table format, including the name, brand/model information, quantity, unit and total weight, physical dimensions, and estimated cost of all electronic components.

**Section 3.3.3 Electrical Design and Cabling:** The selection of cable cross-sections (AWG) based on current-carrying capacity must be demonstrated with calculations. Connector types (XT60, Molex, JST, etc.) selected considering vibration and environmental factors, as well as locking mechanisms, must be specified. Cable weight calculations must be performed, and their effect on the total weight budget must be shown. In addition, EMI/EMC design measures taken to prevent electromagnetic interference must be explained.

**Section 3.3.4 Power System Analysis and Battery Calculations:** For both UAV and UGV separately, a power consumption table including the average and peak current values of all consumer components in the system must be created. In addition, battery capacity (mAh) calculation methods must be provided in detail, considering mission duration (flight/driving time) and safety margins. The suitability of the battery discharge rate (C rating) must be mathematically demonstrated based on the instantaneous maximum current drawn from the system.

**Section 3.3.5 Software Architecture and Image Processing:** This section may be examined under three subheadings:

**Software Infrastructure:** The programming languages (Python, C++, etc.), libraries (OpenCV, ROS, TensorFlow, etc.), and operating systems selected for autonomous systems on the UAV and UGV must be explained together with the reasons for selection and their advantages.

**Image Processing Algorithm:** The working principle of the image processing or artificial intelligence (AI/ML) model used by the UAV for route detection, including dataset preparation, training process, and performance metrics, must be explained and supported with block diagrams.

**Communication Protocol:** The data communication architecture between the UAV and UGV platforms, and the protocols implemented to ensure data integrity and security, must be explained in technical detail.

## **Section 4. Test Activities (10 POINTS)**

In this section, teams are expected to demonstrate the applicability of the conceptual designs submitted in the preliminary evaluation phase by detailing them further, and to present the testing and verification processes carried out for the system and subsystems. The purpose,

implementation method, success criteria, and relationship with system requirements of each test and verification activity must be clearly defined. The content presented under this heading is expected to technically validate the operational capability, reliability, and mission requirement compliance of the developed system. Teams are required to explain test plans, test procedures, and verification methodologies demonstrating that the system meets mission requirements. Acceptance tests performed on system components, integration processes, verification activities, and evaluation methods for test results must be detailed. Visuals and results of conducted acceptance tests must be provided. The schedule for future acceptance tests must also be included. In addition, ground tests that have been carried out must be presented with visuals and results, and the schedule for future ground tests must be provided.

### **Section 5. Requirement Traceability and Evaluation (10 POINTS)**

In this section, teams are expected to systematically present how the requirements listed in the competition requirement set have been addressed throughout the system development process, to what extent they have been fulfilled, and by which verification methods they are managed. Teams must present the distribution of satisfied, under-verification, and unfulfilled requirements using technical data, clearly specify the verification method used for each requirement, and support this information with visual tools such as tables, traceability matrices, graphs, or pie charts. Teams are required to base their verification approach on the Means of Compliance (MoC) methodology. Accordingly, the verification method for each requirement (analysis, test, inspection, demonstration, etc.) must be defined, and the relationship between verification activities and system requirements must be clearly shown. For unmet requirements, the planned verification method, completion approach, and targeted verification process must be specified. The content under this section is expected to demonstrate that teams address requirements management, traceability, and verification processes using a systems engineering approach.